



CHAPTER

11

Dual nature of Radiation and Matter

Sr. No.	Concept	Formulas	Other information
1.	Energy and momentum for photon	Energy of a photon, $E = h\nu = \frac{hc}{\lambda}$. Number of photons emitted per second, $N = \frac{P}{E}$. Momentum of photon, $p = mc = \frac{h\nu}{c} = \frac{h}{\lambda}$. Effective mass of a photon, $m = \frac{h\nu}{c^2}$.	P = power of source c = speed of light. h = Planck's constant ($h = 6.62 \times 10^{-34}$ J-s). e = charge of electron. ν = frequency of photon. λ = wavelength of photon.
2.	Radiation force and radiation pressure	(i) When radiations are falling normally on a perfectly reflecting surface; Force on the surface; $F = \frac{\Delta p}{\Delta t} = \frac{2Nh}{\lambda t} = n \left(\frac{2h}{\lambda} \right)$ $= \left(\frac{P\lambda}{hc} \right) \left(\frac{2h}{\lambda} \right) = \frac{2P}{c}$ Pressure on the surface = $\frac{F}{A} = \frac{2P}{cA} = \frac{2I}{c}$ (ii) Force and pressure on perfectly absorbing surface; $F = \left(\frac{P\lambda}{hc} \right) \left(\frac{h}{\lambda} \right) = \frac{P}{c}$ $\Rightarrow$ Pressure on the surface = $\frac{F}{A} = \frac{P}{Ac} = \frac{I}{c}$	N = total number of photons. t = time. h = Planck's constant. I = intensity of light = $\frac{P}{A}$ A = surface area normal to the light. P = power of the source. n = number of photons per second. c = speed of light. Δp = change in momentum.

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3.	Einstein's photoelectric equation	$K_{\max} = \frac{1}{2}mv_{\max}^2 = E - W_0 \quad [W_0 = h\nu_0]$ $\Rightarrow \frac{1}{2}mv_{\max}^2 = h\nu - W_0 = h(\nu - \nu_0) = h\left[\frac{c}{\lambda} - \frac{c}{\lambda_0}\right]$ If V_0 is the stopping potential, the maximum kinetic energy of the ejected photo electrons, $K = \frac{1}{2}mv_{\max}^2 = eV_0$	E = energy of photon ν_0 = threshold frequency. $K_{\max}$ = maximum kinetic energy ν = frequency of the incident radiation W_0 = work function of metal. V_0 = stopping potential
4.	De-Broglie Waves	Kinetic energy, $K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ $\therefore$ Momentum, $p = \sqrt{2mK}$ de – Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}$. de-Broglie wavelength of an electron beam accelerated through a potential difference of V volts: $\lambda = \frac{h}{\sqrt{2meV}} = \frac{1.227}{\sqrt{V}} \text{ nm}$	m = mass of the particle. V = potential difference. v = velocity of particle. λ = de-Broglie wavelength of particle. p = momentum of particle.